

# Experiment 8: Create and Manage a Graph Database

|                  |                                  |                  |            |
|------------------|----------------------------------|------------------|------------|
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| Roll No(s):      | 36, 38, 39, 40                   | Experiment Date: | 2026-09-19 |
| Quiz Evaluation: | 6 / 10 (60%)                     |                  |            |

## 1. Aim & Expected Outcome

Aim: Install Neo4j, create nodes and relationships, and perform basic graph operations.  
Expected outcome: A functioning graph database containing connected entities.

## 2. Learning Objectives

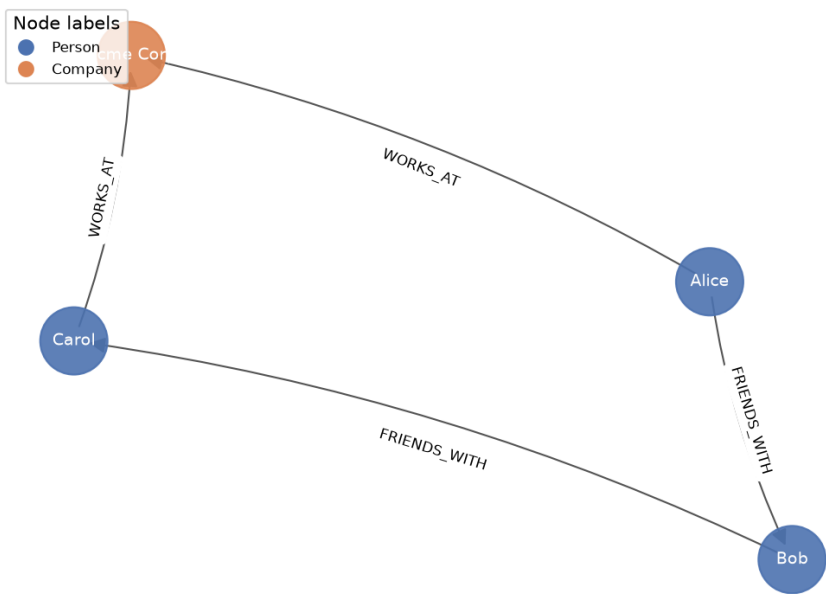
- Understand what a graph database is and how it differs from a relational database.
- Learn the building blocks of a property graph: nodes, labels, relationships and properties.
- Create, read, update and delete (CRUD) nodes and relationships.
- Query the graph for neighbors and shortest paths between entities.
- Visualize the resulting graph and record observations in a lab report.

## 3. Recorded Experimental Trials & Data

| Trial | Operation         | Nodes | Rels | Result                                         | Time     |
|-------|-------------------|-------|------|------------------------------------------------|----------|
| 1     | LOAD sample graph | 4     | 4    | Loaded sample graph (4 nodes, 4 relationships) | 10:37:39 |

## 4. Final Graph State

Nodes: 4   Relationships: 4



- Alice -[FRIENDS\_WITH]-> Bob
- Bob -[FRIENDS\_WITH]-> Carol
- Alice -[WORKS\_AT]-> Acme Corp
- Carol -[WORKS\_AT]-> Acme Corp

## 5. Observations & Analysis

The experimental trials demonstrated the creation and management of a graph database of connected entities, matching the expected outcome.

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*Instructor / Student Signature*